VF-B20 Seam Stitch Tear-Soling

Test Method for Seam Stitch Tear- Soling	**Test Method #: VF- B20** **REF #: TM-R- 036**
Date Issued: August 12, 2024 **Revision #: 2**	**Reason for revision: Update references from SATRA TM71 to SATRA TM5**

PURPOSE AND SCOPE: To determine the ability of rubber soling material to hold stitches or clenched fastenings.

REFERENCES: SATRA TM 5

EQUIPMENT AND MATERIAL:

1. Tensile testing machine
2. Clamping jaw *(Refer to sample preparation)*
3. Steel wire of diameter 0.90 + 0.02 / - 0.01 mm and minimum length of 150 mm.
4. A drilling jig *(Refer to sample preparation*
5. Caliper or thickness gauge
6. Drill bit of diameter 1.60 +0.02 / - 0.01 mm
7. Electric drill
8. A mandrel of diameter 7 ± 0.5mm to form the wire into a loop
9. Rubber Soling

SAMPLE PREPARATION:

Condition the sample for at least 24 hours at the standard laboratory condition at 23.0 ± 2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Features of the following auxiliaries:

- Clamping jaw : the fixture shall consist of two parts:
  - A flat horizontal plate in the center of which is a rectangular aperture 12.0 ± 0.5mm by 6.0 ± 0.5mm through which the wire loop can be drawn.
  - A means of clamping the wire in a loop so that the legs of the loop are parallel and vertical, 8mm apart, and equidistant from the lines of force during the test.

- A drilling jig: The jig shall allow three sets of holes to be drilled in the test specimen. Each pair of holes shall be 8mm apart on center. Between adjacent pairs there shall be a minimum distance of 11mm between the centers of the two adjacent holes.

2. When possible cut two rectangular test specimens approximately 75mm x 25mm from the test sheet. Cut one from the principal direction and the second 90° to this in the cross direction. When the material is such that two specimens cannot be cut, one specimen shall be sufficient. For solid rubber the specimen should be 3.2mm thick. For cellular rubbers the specimen should be 4.8mm thick with the solid skin removed.

3. Mark the principal direction on each specimen.

4. Place one of the test specimens into the drilling jig and use the drill and bit to drill six holes through the specimen. Repeat this for the remaining test specimens.

5. Measure and record the thickness of the test specimen at three locations corresponding to the center of the three pairs of holes.

6. Calculate the arithmetic mean of the three thickness measurements, **T (cm**)

REPETITONS: Test three (3) samples

TEST PROCEDURE:

1. Fix the two parts of the clamp into the tensile testing machine.
2. Using the mandrel, form the wire into a smooth semi-circular loop.
3. Thread the wire loop through a pair of holes in the test specimen from the face side. In the case of insole materials, the face side is usually the foot side.
4. Place the test specimen onto the flat surface of the upper clamp so that the wire loop ends protrude down through the rectangular aperture. For very stretchy rubber it may be necessary to use a block with a smaller aperture, which is should then be placed on top of the larger aperture.
5. Firmly clamp both free ends of the wire loop into the lower clamp ensuring that the legs of the loop remain parallel and vertical.
6. Operate the tensile testing machine with a jaw separation rate of 100 ± 20 mm/min and record the maximum force required to pull the loop through the test specimen, **F (kgf)**
7. Repeat the procedure in sections 3.9 through 3.12 on the remaining two pairs of holes in the test specimen.

CALCULATION:

1. Calculate the arithmetic mean of the three maximum forces recorded to determine the stitch tear force of the material : F (kgf)
2. Get the average thickness : T (cm)
3. Stitch tear force (kgf/cm) : F / T

REPORTS:

1. Stitch tear = kgf/cm
2. The size of the aperture used
3. Any deviations from this procedure.